Three Multiplier Channels and a Lefschetz Identity for the Full Multiplication Lattès Family

Route-A structural certificate HCS-C180

Abstract

Let an integer multiplication map $[m]$ on a complex elliptic curve descend through the quotient by sign to a Lattès map f_m on the sphere. Uniformly over every elliptic modulus, $m \geq 2$, and iterate $n \geq 1$, we resolve the fixed set into regular plus, regular minus, and branch channels. With $a = m^n$ and $h = 1$ for even a, $h = 4$ for odd a, their counts are $((a-1)^2 - h)/2$, $((a+1)^2 - h)/2$, and h, with multipliers $a, -a, a^2$. They give a holomorphic Lefschetz sum equal to one. We derive the exact period ledger, $\zeta_{\mathrm{AM}} = ((1-z)(1-m^2z))^{-1}$, and the Wold model $I_{\mathbb{C}} \oplus S^{(\aleph_0)}$ of the natural Haar Koopman isometry. The theorem is a full Lattès classification but a sharp Route-A stop: its counts are modulus-blind and its canonical operator is noncompact.

Keywords: Lattès map; elliptic torsion; holomorphic Lefschetz formula; Artin–Mazur zeta; Wold decomposition.

中文摘要

对任意复椭圆曲线、整数 $m \geq 2$ 与迭代次数 $n \geq 1$，本文完整分类乘法 $[m]$ 经 $\{\pm 1\}$ 商降下的拉泰斯映射不动点。令 $a = m^n$；不动类分为 正、负与分支三通道，其乘子分别为 $a, -a, a^2$，计数中的分支修正由 a 的 奇偶性决定。三通道给出恰为一的莱夫谢茨和，并导出精确周期、动力学 泽塔与自然哈尔及库普曼等距算子的沃尔德分解。该结果是完整动力学定理，但因计数对模参数失明且算子非紧，它同时构成严格的甲路线停止定理。

1 Family and fixed-class theorem

Let $E = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ be any complex elliptic curve, $\pi : E \to E/\{\pm 1\} \simeq \mathbb{P}^1$, and $m \geq 2$. Define f_m by $f_m \pi = \pi[m]$. This classical Lattès construction is prior work; see Milnor [1]. We claim no priority for that construction, torsion periodic points, or the holomorphic Lefschetz formula.

Fix $n \geq 1$, put $a = m^n$, and define $h(a) = 1$ when a is even and $h(a) = 4$ when a is odd.

Theorem 1 (Three-channel census). *For every E, m, n,*

$$\mathrm{Fix}(f_m^n) = \big(E[a-1] \cup E[a+1]\big)/\{\pm 1\}.$$

The fixed classes split disjointly as

channel	*count*	*multiplier*	*origin*
regular plus	$N_+ = ((a-1)^2 - h)/2$	$+a$	$E[a-1]$
regular minus	$N_- = ((a+1)^2 - h)/2$	$-a$	$E[a+1]$
branch	$N_{\mathrm{br}} = h$	a^2	*common two-torsion*

In particular, $\#\,\mathrm{Fix}(f_m^n) = a^2 + 1$ *and*

$$\frac{N_+}{1-a} + \frac{N_-}{1+a} + \frac{N_{\mathrm{br}}}{1-a^2} = 1. \tag{1}$$

Proof. The equality $f_m^n \pi(P) = \pi(P)$ is equivalent to $[a]P = P$ or $[a]P = -P$, which gives the torsion union. Its intersection is $E[\gcd(a-1, a+1)]$: it has one element for even a and four for odd a. These are exactly the quotient branch classes. Remove the intersection and pair the remaining points by sign to obtain the counts.

Off the branch locus, π is locally biholomorphic. The chain rule gives $+a$ on the first torsion set. On the second, the identity $D\pi_{-P} = -D\pi_P$ supplies the sign $-a$. At two-torsion a quotient coordinate is the square of a torus coordinate, so the multiplier is a^2. The total and (1) follow by direct algebra. ☐

Why the branch channel cannot be merged

The common torsion classes are not double-counted regular points. Their local coordinate is quadratic, so replacing them by either regular channel would assign multiplier $\pm a$ instead of a^2. The resulting fixed total could still be repaired arithmetically, but the local Lefschetz sum would be wrong. Thus the three-channel statement is strictly stronger than the scalar count $a^2 + 1$ and is the minimal branch-aware invariant of the quotient.

2 Exact periods and zeta

Corollary 2. *The exact-period point count and primitive-cycle count are*

$$P_m(n) = \sum_{d|n} \mu(n/d)(m^{2d}+1), \qquad P_m(n)/n,$$

and

$$\zeta_{\mathrm{AM}}(z) = \exp\left(\sum_{n\geq 1}(m^{2n}+1)\frac{z^n}{n}\right) = \frac{1}{(1-z)(1-m^2z)}. \tag{2}$$

Proof. Möbius inversion separates exact periods, and exact-period points partition into cycles of size n. Equation (2) is the sum of two geometric logarithmic series. □

3 Natural operator and Wold boundary

Let $\nu = \pi_* \operatorname{Haar}_E$ and let $U_m g = g \circ f_m$ on $L^2(\mathbb{P}^1, \nu)$. Pullback identifies this Hilbert space with the even subspace of $L^2(E)$. For nonzero dual-lattice k, set $c_k = (e_k + e_{-k})/\sqrt{2}$, indexed modulo sign. Then

$$U_m c_k = c_{mk}.$$

Every nonzero $k \in \mathbb{Z}^2$ is uniquely $m^j r$ with $r \notin m\mathbb{Z}^2$, modulo sign. Each primitive r starts a unilateral shift chain, and there are countably many such roots. Constants form the only unitary summand. Therefore

$$U_m \simeq I_{\mathbb{C}} \oplus S^{(\aleph_0)}, \qquad U_m^* c_k = \begin{cases} c_{k/m}, & k \in m\mathbb{Z}^2, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

The isometry is proper, noncompact, and outside every finite Schatten class; an ordinary Fredholm determinant $\det(I - zU_m)$ is unavailable. Pullback of an ample line bundle is only a formal alternative: $[m]^*L$ has degree $m^2 \deg L$, so it does not act on one fixed quantization space.

This operator statement uses only pushed-forward Haar measure. It neither chooses a polarization nor identifies Hilbert spaces of different line-bundle degree. Consequently the Wold model is canonical for the frozen measurable system, whereas the line-bundle suggestion is deliberately recorded only as a change-of-space hint.

4 Exact validation ledger

The proof above is global; finite computations are regression sentinels, not proof. Exact software independently enumerated the following frozen ledgers.

ledger	exact total
formula rows ($2 \leq m \leq 10,\ 1 \leq n \leq 12$)	108
direct torsion-union enumerations	16
materialized rational torus points	25,290
even Fourier-mode rows	5,880
independent-checker assertions	43,184
SymPy identity checks	18,065
repaired/stale-hash mutation rejections	$23+1$

The torsion sentinel forms both finite rational tori as sets before taking their intersection and sign quotient. The checker imports no producer code; byte replay is exact. This distinguishes branch-class validation from mere substitution into the closed count formula.

5 Route-A decision

The exact tuple is

$$(\texttt{A0\_FAIL}, \texttt{A1\_WEAK}, \texttt{A2\_FAIL}, \texttt{A3\_FAIL}, \texttt{A4\_FORMAL\_HINT}).$$

A0 fails because torsion classes have no intrinsic rational-prime labels or prime-power weights. A1 remains weak despite complete orbit counts because no arithmetic labels or target amplitudes occur. The elementary zeta is independent of τ, giving A2 and A3 failure. Equation (3) is canonical but non-Schatten, so A4 is only a formal hint. A0 failure forces `ROUTE_A_REJECTED`.

gate	decisive boundary
A0_FAIL	no rational-prime labels, logarithmic-prime clock, or prime-power weights
A1_WEAK	complete primitive ledger, but no arithmetic labels or target amplitudes
A2_FAIL	exact elementary zeta is independent of elliptic modulus
A3_FAIL	rational continuation has no target functional equation or Weil compression
A4_FORMAL_HINT	canonical Koopman lift is a proper non-Schatten isometry

Limitations and ownership

The theorem concerns the multiplication family descended through the sign quotient. It does not cover translated affine maps, general isogenies between different elliptic curves, or arbitrary postcritically finite rational maps. The disappearance of τ from the fixed count is a proved universality statement, not evidence for an arithmetic target hidden in the modulus.

The Wold theorem is an observable-space classification. It supplies neither a self-adjoint generator nor an ordinary Fredholm determinant, and changing to a natural extension or a varying line-bundle space would change the frozen candidate. These alternatives are therefore boundaries rather than repairs.

Milnor's work is cited for the classical Lattès setting. The branch-resolved ledger, Route-A audit, and software integrity artifacts are the contribution of this certificate; no general literature-priority claim is made for the component identities. Finite ledgers can expose implementation errors but cannot establish the all-parameter theorem.

The scope is NO_BAD_EULER_OR_ROOT_NUMBER. No prime table, target divisor, arithmetic local factor, Euler factor, root number, automorphy, Hilbert–Pólya operator, Route-B authorization, external review, or acceptance rate is claimed.

References

[1] J. Milnor, *On Lattès Maps*, arXiv:math/0402147 (2004), https://arxiv.org/abs/math/0402147.